

Day 14 - Help Session



Announcements

- HW 4 is due today
 - We will work HW 4 Exercise 4 together + Q&A
 - Office Hours (14:00-16:00; 1248 BPS)
- Midterm 1 is available now
 - We will talk about it Monday (come with questions)
 - There will be no office hours on Oct 3rd (DC traveling)

Exercise 4, Example of potential

Consider a particle of mass m moving according to the potential:

$$V(x, y, z) = A \exp \left\{ -\frac{x^2 + z^2}{2a^2} \right\}.$$

We can think of this potential as the energy landscape of a particle in three dimensions. That is, you can imagine a particle moving around this potential like a ball rolling around a landscape. That analogy is not perfect, but it is a good way to help us think about stability and equilibrium.

Exercise 4, Example of potential

$$V(x, y, z) = A \exp \left\{ -\frac{x^2 + z^2}{2a^2} \right\}.$$

- 4a Plot this potential or sketch a plot of it. You can use perspective plots, contour plots or any other plot you find useful.
- 4b What are some feature you notice with this potential? What happens when you change A and a ?

Exercise 4, Example of potential

$$V(x, y, z) = A \exp \left\{ -\frac{x^2 + z^2}{2a^2} \right\}.$$

- 4c Imagine a particle moving in this potential, what are some expected trajectories?
- 4d Do there appear to be any equilibrium points? If so, are they stable or unstable?
- 4a Is the resulting force conservative? Why?